Correct-Response Novelty in Full-Weight LLM Evolution

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Abstract

We test an adaptation of historical response novelty to full-weight Gaussian evolution strategies on witnessed arithmetic problems. The comparison uses one paired training seed and an untrained base model. No completed paired primary endpoint is available. Problem-resampling intervals exclude training-seed variability. All tables are generated from persisted evaluations; incomplete outcomes are labeled.

Context, motivation and introduction

The experiment asks whether exploration among correct answer forms improves held-out task success. ExTra [1] supplies the historical semantic novelty concept; ES at Scale [2] supplies the full-weight optimization reference. Their combination changes the optimizer, sibling population, task and normalization, so it is an adaptation rather than a reproduction. A reward-only ES control is necessary because ES can itself change solution coverage [3].

Background and related work

Novelty-seeking ES was previously studied in control [4]. LLM-specific alternatives include EVOL-RL's within-group novelty [5], DARLING's quality/diversity objective [6], and the model/task coevolution of AC/DC [7]. These use different training or population mechanisms. The repository literature audit distinguishes prompt evolution, artifact search and actual model-parameter training, with code/license verification.

Method

The model is Qwen/Qwen2.5-0.5B-Instruct, revision 7ae557604adf67be50417f59c2c2f167def9a775. Each generation evaluates 30 independent Gaussian candidates (sigma 0.001) on 32 rotating training problems and 16 fixed training probes. Mean task reward is combined with 0.25 times correctness-gated cosine novelty for the main condition. The archive retains 6 correct embeddings per probe after seeded insertion. Average centered fitness ranks weight noise in the update; effective alpha is 0.0005. FP32 masters reconstruct BF16 candidates afresh and tied tensors are updated once. Prefix regeneration and backpropagation are absent.

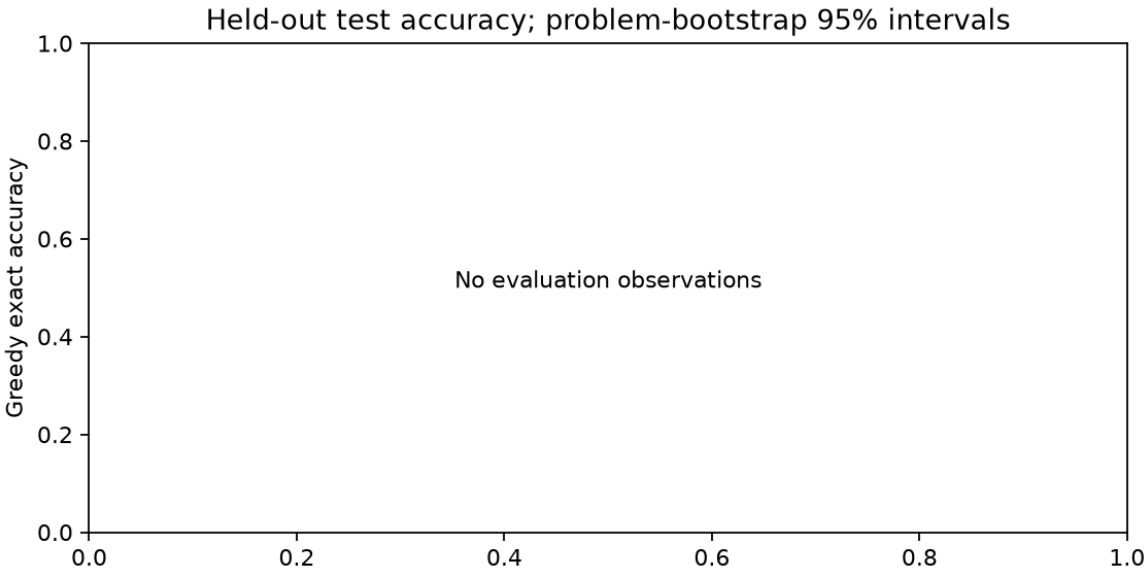
Data, evaluation and preregistered analysis

Synthetic problems have exact rational witnesses. Global number-multiset isolation prevents permutations or alternative targets crossing splits. Only constants supplied in the prompt and allowlisted arithmetic are accepted. Greedy test accuracy at the final scheduled checkpoint is primary. The directional hypothesis is falsified for this pilot if main minus control accuracy is nonpositive. A paired problem bootstrap quantifies conditional uncertainty. Sampled pass@k follows the unbiased subset estimator [8]. Validation cannot select checkpoints. Exact dataset hashes, schedules, timing rules and sampling settings are recorded in reports/preregistration.md and configs/pilot.json.

Results and claims

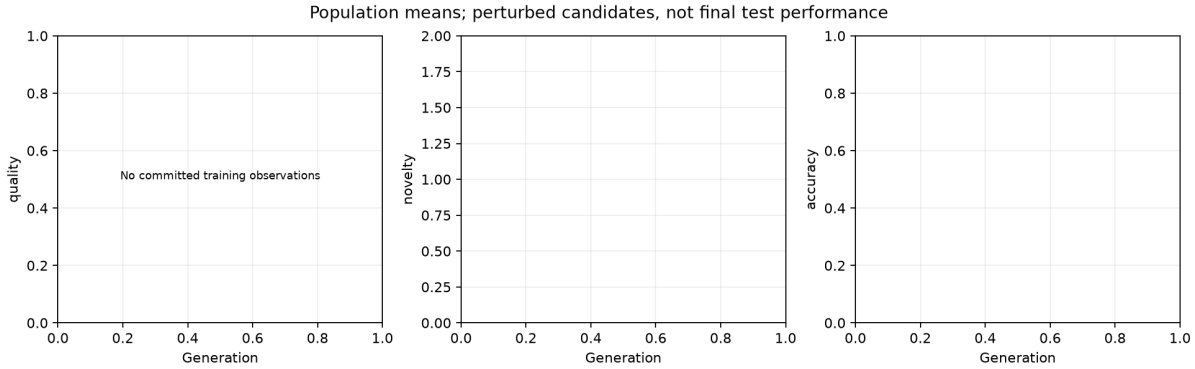
No completed paired primary endpoint is available.

Condition	Test accuracy	95% interval	Hard accuracy	pass@1	pass@8	Status


Held-out test accuracy; problem-bootstrap 95% intervals
Greedy exact accuracy
No evaluation observations
0.0
0.2
0.4
0.6
0.8
1.0

Training dynamics and output inspection

Population metrics describe perturbed candidates, not the master checkpoint. Reward is the optimization signal; no cross-entropy loss is optimized in this pilot. The interactive result page shows the same first test problems across methods, including failures, without choosing examples by outcome.



Implications and discussion

This pilot is a controlled objective intervention under one initialization/seed schedule. It can identify a conditional difference or a feasibility failure, but cannot establish a general advantage of novelty training. Any improvement must be checked across additional seeds and separated from probe memorization. A rise in novelty alone, or a selected qualitative example, is insufficient evidence.

Limitations

Training-seed sensitivity is unmeasured. Embedding similarity is a proxy and may reward irrelevant wording; canonical expression trees are an independent syntactic diagnostic, not full algebraic equivalence. Synthetic arithmetic does not measure general language quality. Similar pretraining patterns cannot be excluded. The fixed-probe objective can overfit. Small time budgets, laptop thermal variability and batch-level floating-point behavior limit extrapolation. Secondary endpoints are descriptive; no multiple-comparison significance claims are made.

Conclusion

No completed paired primary endpoint is available. Multi-seed studies and archive ablations require a separately frozen amendment after this result is published.

Appendix: reproducibility and failures

Run uv sync --locked, uv run pytest, uv run novelty-es prepare, uv run novelty-es download, and uv run novelty-es run. Repeating run resumes the same clock. uv run novelty-es report regenerates tables, figures, this paper and the static website. Atomic checkpoints contain weights, all RNG streams, archive, cursor, config and deadline. Recorded technical failures: 0. Experiment elapsed seconds: 0.000. Exact exceptions and incomplete conditions are retained in the generated Markdown report and JSON artifacts.

References

1. ExTra

2. Evolution Strategies at Scale

3. Beyond the Best Guess

4. Novelty-seeking ES

5. EVOL-RL

6. DARLING

7. AC/DC

8. Evaluating Large Language Models Trained on Code

Research protocol and source

AI statement

Codex wrote this codebase and paper, and human cleanup and rewriting are in progress. A human provided the exact specification that was executed.